



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

Presentation

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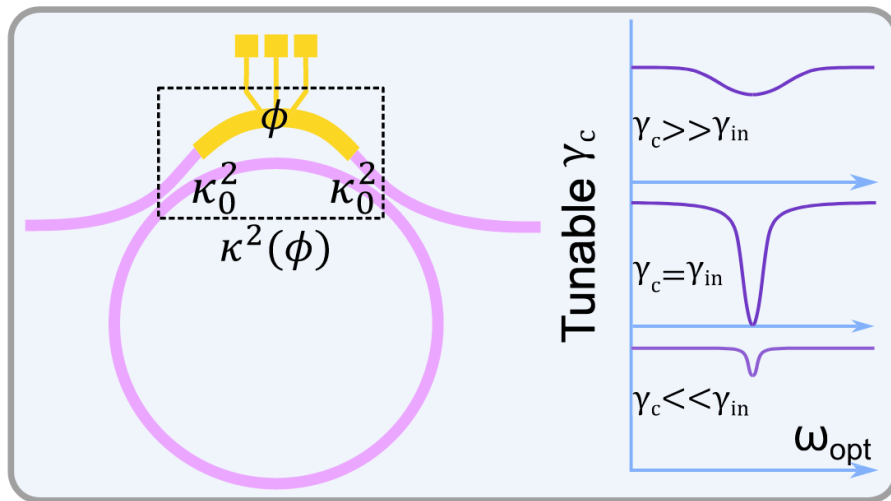
6/4/2026



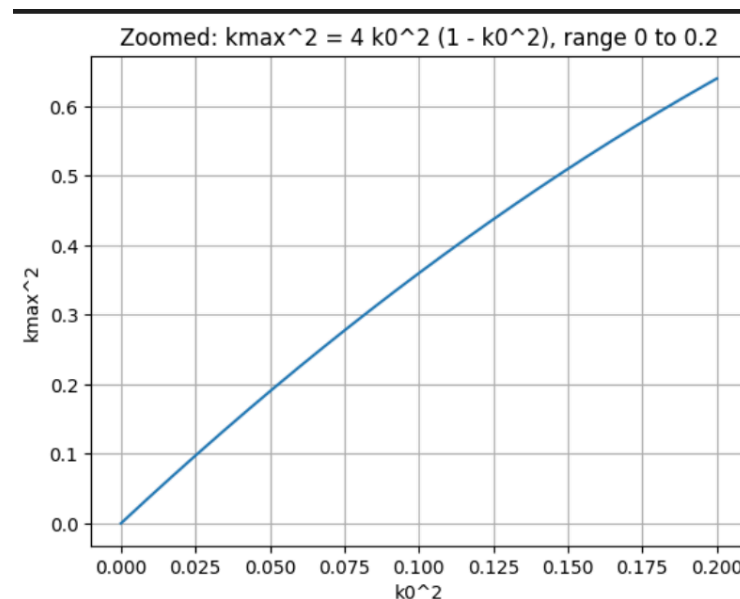


Coupling tunable ring

Two-point-coupled ultra-high Q resonator



$$\kappa_{\max}^2 = 4\kappa_0^2(1 - \kappa_0^2)$$



pt

Coupling tunable ring

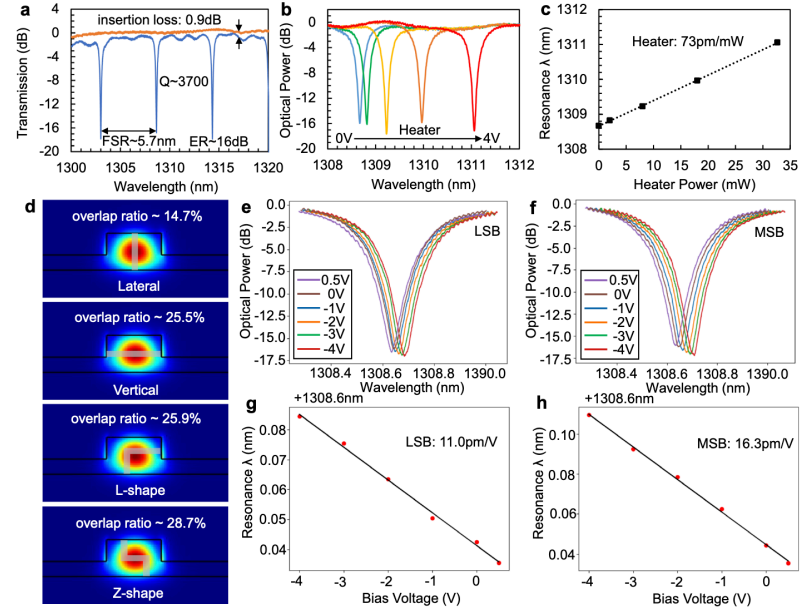
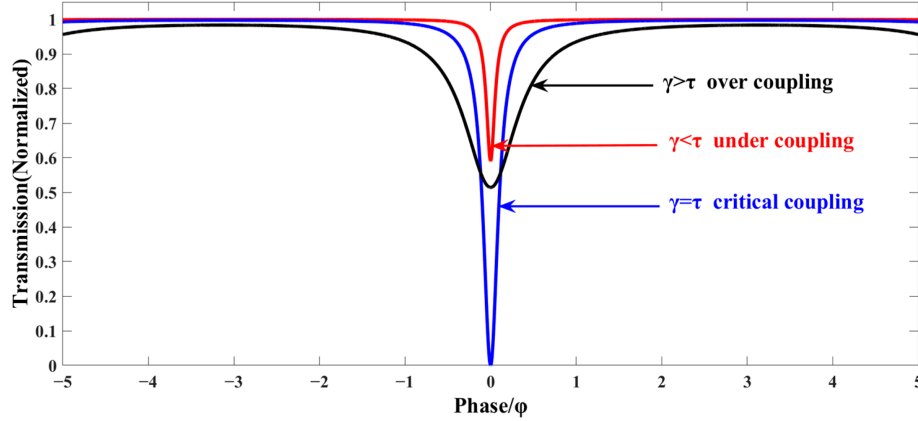


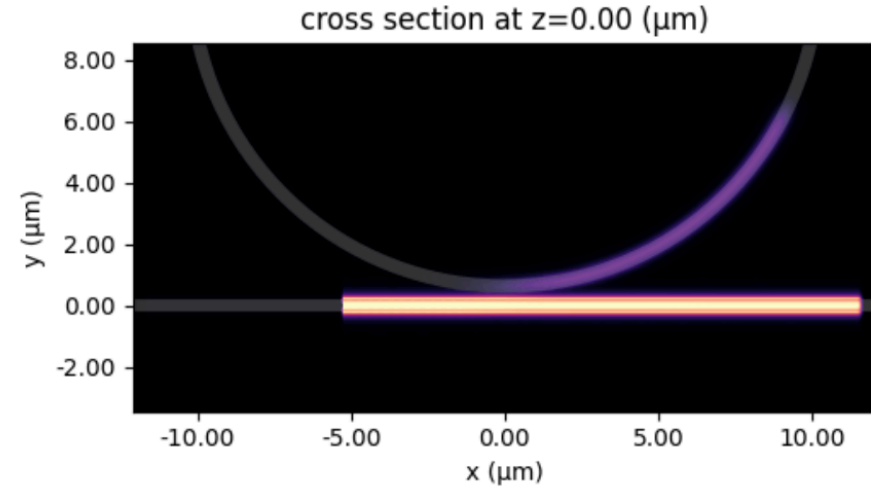
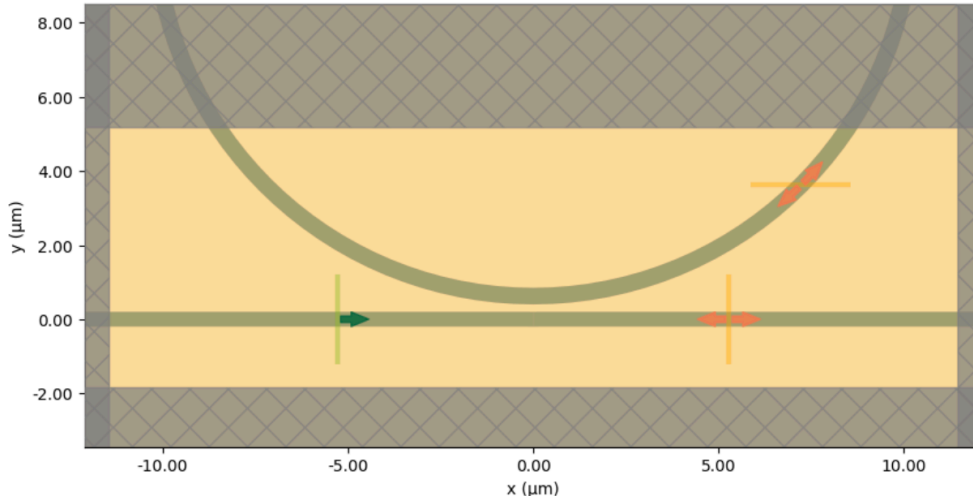
Fig. 2 | Direct current (DC) characteristics of a single microring modulator (MRM). **a**, Measured transmission spectrum of a single-channel MRM (blue) and a pair of grating couplers (orange). **b**, Measured transmission spectrum versus heater voltage ranging from 0 V to 4 V. **c**, Resonance wavelength versus heater power. **d**, Calculated overlap ratios between the waveguide transverse electric (TE) mode and depletion regions of different junctions. **e**, **f**, Measured transmission spectrum versus junction voltage from 0.5 V to 4 V at the least significant bit (LSB) and most significant bit (MSB). **g**, **h**, Extracted resonance wavelengths versus junction voltage for LSB and MSB.

Over coupling can give a moderate slope

$$|\kappa|^2 \approx 20\% \quad \tau_{ph} \approx \frac{\lambda Q}{2\pi c} = \frac{Q}{\omega_0} \quad f_{3dB} \approx \frac{1}{2\pi\tau}$$



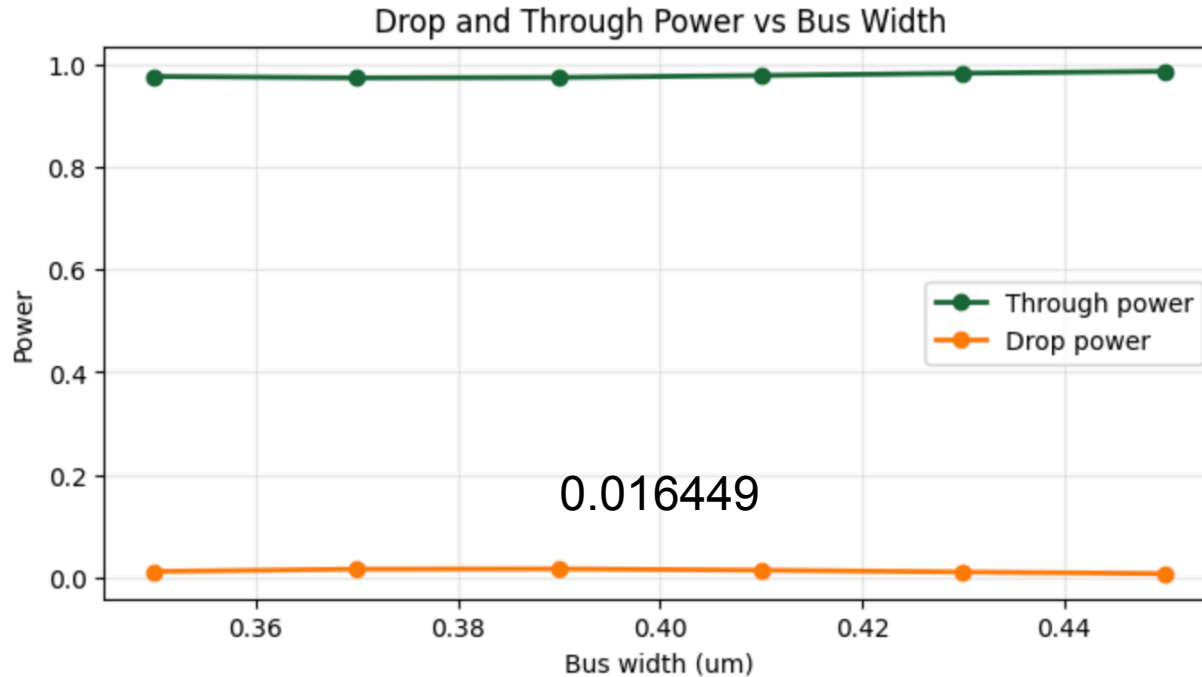
Coupling tunable ring



Gap 0.2 μm (foundry limitation) transmission: 0.923654 drop: 0.067563
Gap 0.3 μm transmission: 0.976310 drop: 0.015414



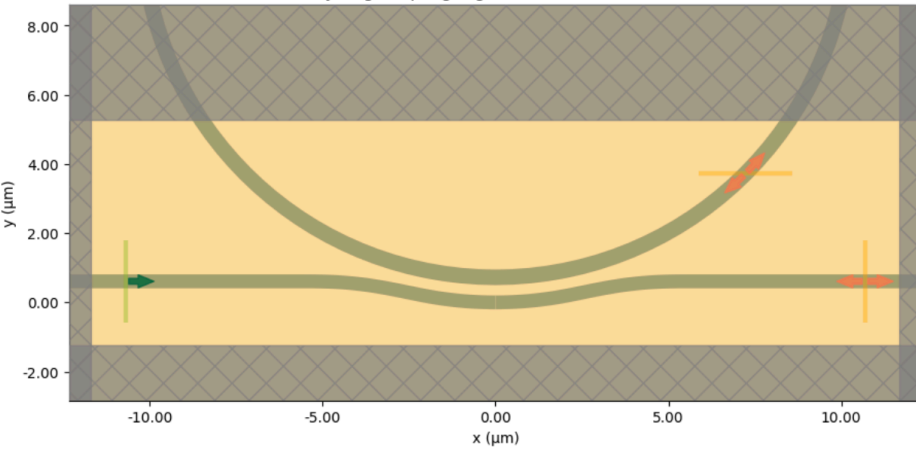
Coupling tunable ring



Gap: 0.3um coupling angle: 0 deg (point coupling)

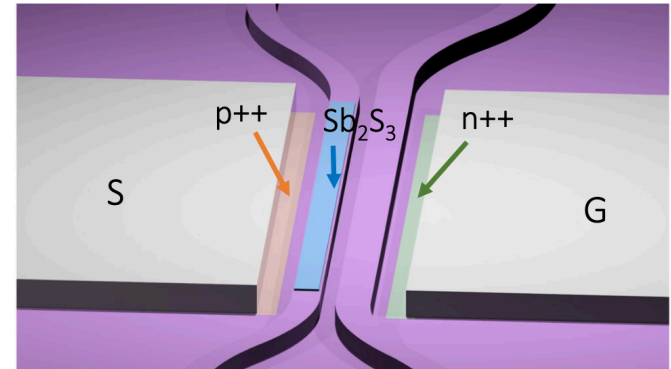


Coupling tunable ring



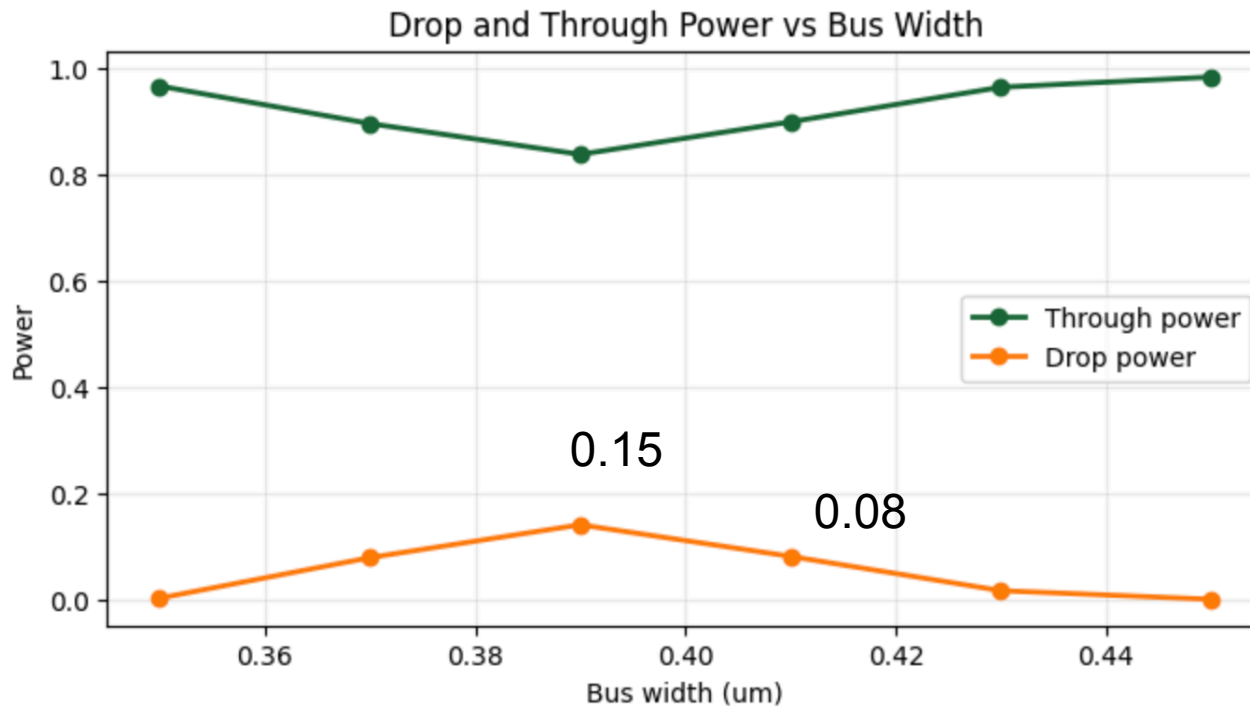
Radius: 10 μm
Ring width: 0.45 μm
Bus width: 0.4 μm

a





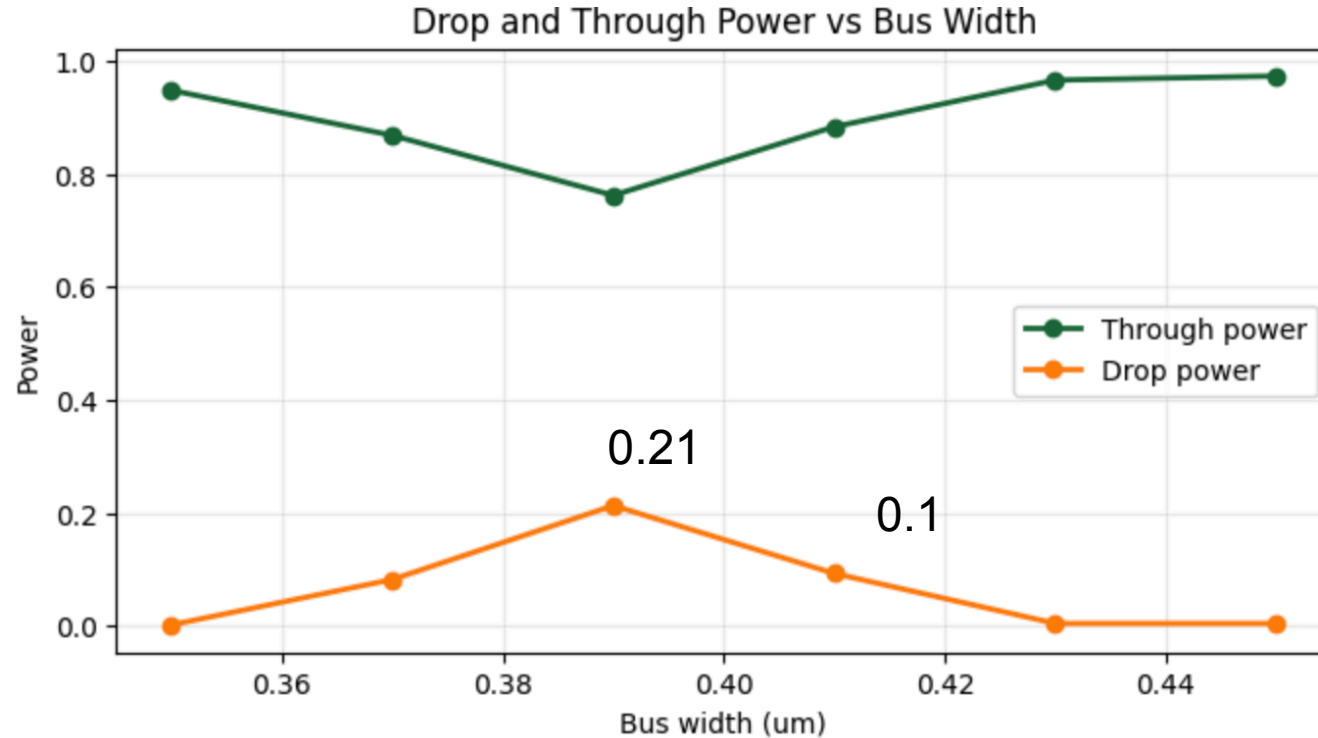
Coupling tunable ring



Gap: 0.3um coupling angle: 30 deg

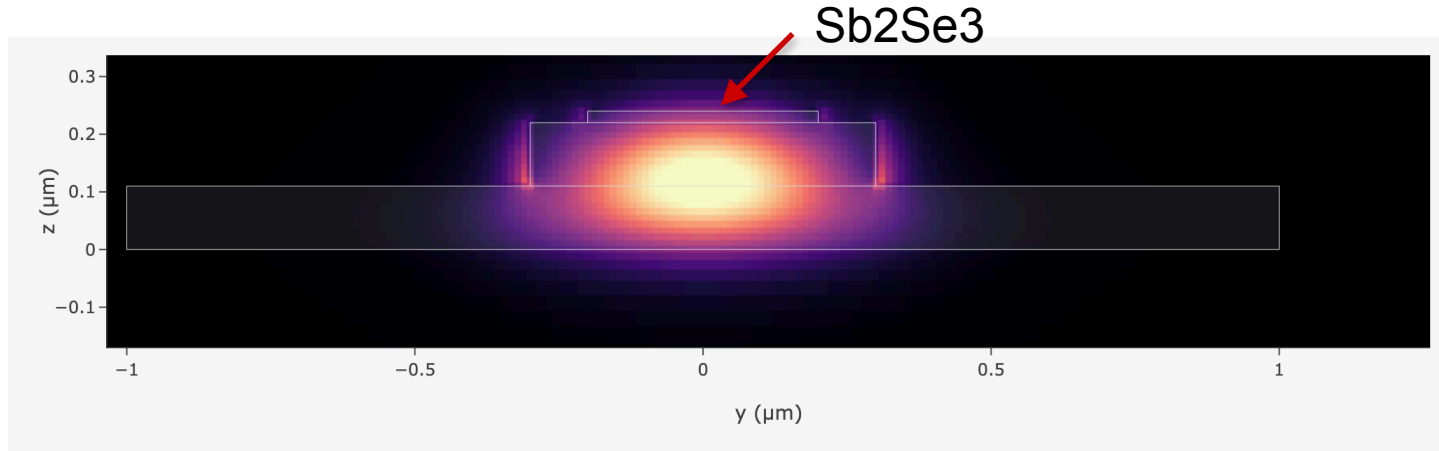


Coupling tunable ring



Gap: 0.3um coupling angle: 40 deg

Coupling tunable ring



Refractive index of Sb_2Se_3

Crystalline: 4.050

Amorphous: 3.285

Effective index

Pure silicon: 2.660114

C-PCM: 2.725577

A-PCM: 2.698020

$\Delta n = 0.027$ *possible* $L_\pi = 28.7 \mu\text{m}$



Coupling tunable ring



Coupling tunable ring